

1st Question:

Code:

```
import pandas as pd
```

```
# Read the Excel file
```

```
df = pd.read_excel("one.xlsx")
```

```
# Display the dataset
```

```
print("Departments Dataset:")
```

```
print(df)
```

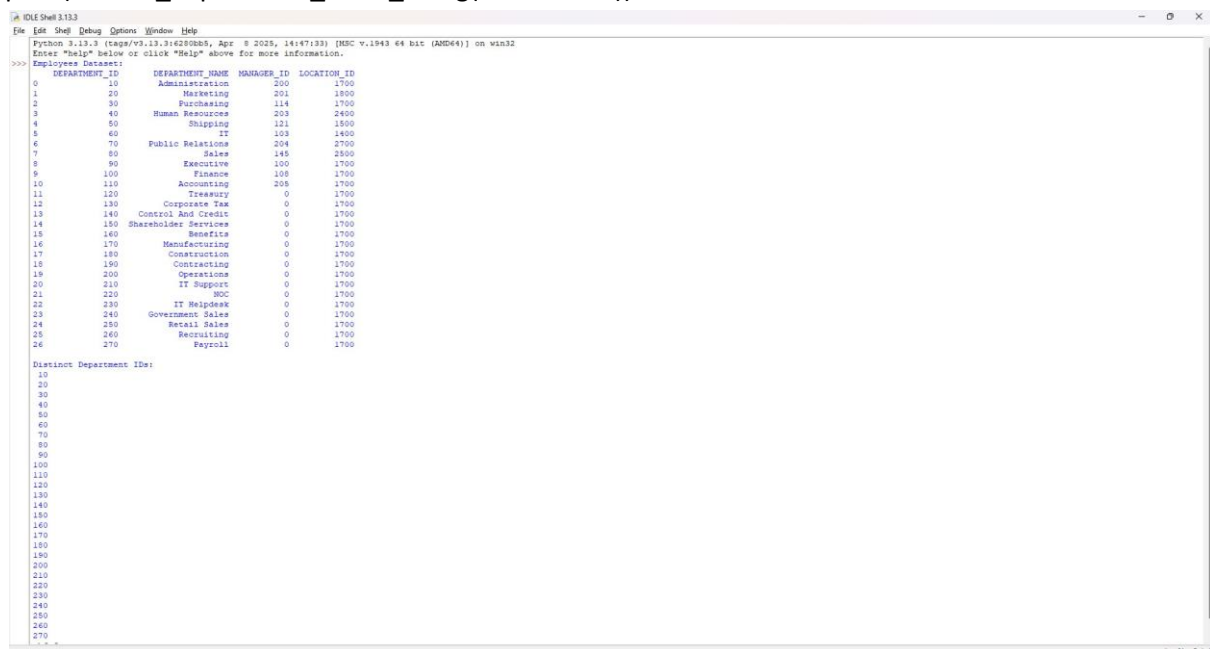
```
# Select distinct department IDs
```

```
distinct_department_ids = df["DEPARTMENT_ID"].drop_duplicates()
```

```
# Display the result
```

```
print("\nDistinct Department IDs:")
```

```
print(distinct_department_ids.to_string(index=False))
```



```
Python 3.13.3 (tags/v3.13.3:4280005, Apr 8 2025, 14:47:13) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
Employees Dataset:
DEPARTMENT_ID  DEPARTMENT_NAME  MANAGER_ID  LOCATION_ID
0              10      Administration      200         1700
1              20      Marketing          201         1800
2              30      Purchasing         114         1700
3              40      Human Resources     203         2400
4              50      Shipping          121         1800
5              60      IT              103         1400
6              70      Public Relations  204         2700
7              80      Sales           145         2500
8              90      Executive       100         1700
9             100      Finance           108         1700
10            110      Accounting        205         1700
11            120      Treasury            0         1700
12            130      Corporate Tax      0         1700
13            140      Control And Credit  0         1700
14            150      Shareholder Services  0         1700
15            160      Benefits            0         1700
16            170      Manufacturing      0         1700
17            180      Construction      0         1700
18            190      Contracting        0         1700
19            200      Operations          0         1700
20            210      IT Support          0         1700
21            220      SOC              0         1700
22            230      IT Helpdesk        0         1700
23            240      Government Sales  0         1700
24            250      Retail Sales      0         1700
25            260      Recruiting        0         1700
26            270      Payroll           0         1700

Distinct Department IDs:
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150
160
170
180
190
200
210
220
230
240
250
260
270
```

2nd Question:

Code:

```
import pandas as pd
```

```
# Read Excel file
```

```
df = pd.read_excel("history.xlsx")
```

```
print("Employees History")
```

```
print(df)
```

```
# Employees who have done two or more jobs
```

```
result = df["EMPLOYEE_ID"].value_counts()
```

```
print("\nEmployees who have done two or more jobs:")
```

```
print(result[result >= 2].index.to_list())
```

```
===== RESTART: D:/query/exp 2.py =====
Employees History
  EMPLOYEE_ID  START DATE  END DATE  JOB_ID  DEPARTMENT_ID
0         102  2001-01-13  2006-07-24   IT_PROG             60
1         101  1997-09-21  2001-10-27  AC_ACCOUNT             110
2         101  2001-10-28  2005-03-15   AC_MGR             110
3         201  2004-02-17  2007-12-19   MK_REP             20
4        114  2006-03-24  2007-12-31  ST_CLERK             50
5        122  2007-01-01  2007-12-31  ST_CLERK             50
6         200  1995-09-17  2001-06-17  AD_ASST             90
7         176  2006-03-24  2006-12-31   SA_REP             80
8         176  2007-01-01  2007-12-31   SA_MAN             80
9         200  2002-07-01  2006-12-31  AC_ACCOUNT             90

Employees who have done two or more jobs:
[101, 200, 176]
>|
```

3rd question:

Code:

```
import pandas as pd
```

```
# Read Excel file
```

```
df = pd.read_excel("jobs.xlsx")
```

```
print("Jobs Dataset")
```

```
print(df)
```

```
# Sort by JOB_TITLE in descending order
```

```
result = df.sort_values(by="JOB_TITLE", ascending=False)
```

```
print("\nJobs in Descending Order of Job Title")
```

```
print(result)
```

Screenshot output:

```
===== RESTART: D:/query/exp 3.py =====
Jobs Dataset
   JOB_ID JOB_TITLE MIN_SALARY MAX_SALARY
0  AD_PRES  President    20080    40000
1    AD_VP Administration Vice President  15000    30000
2  AD_ASST Administration Assistant      3000     6000
3   FI_MGR   Finance Manager      8200    16000
4 FI_ACCOUNT      Accountant      4200     9000
5   AC_MGR Accounting Manager      8200    16000
6 AC_ACCOUNT Public Accountant      4200     9000
7   SA_MAN   Sales Manager     10000    20080
8   SA_REP Sales Representative      6000    12008
9   PU_MAN Purchasing Manager      8000    15000
10  PU_CLERK Purchasing Clerk       2500     5500
11   ST_MAN   Stock Manager       5500     8500
12  ST_CLERK   Stock Clerk        2008     5000
13  SH_CLERK Shipping Clerk       2500     5500
14   IT_PROG Programmer         4000    10000
15  MK_MAN Marketing Manager       9000    15000
16  MK_REP Marketing Representative      4000     9000
17  HR_REP Human Resources Representative      4000     9000
18  PR_REP Public Relations Representative      4500    10500

Jobs in Descending Order of Job Title
   JOB_ID JOB_TITLE MIN_SALARY MAX_SALARY
11   ST_MAN   Stock Manager       5500     8500
12  ST_CLERK   Stock Clerk        2008     5000
13  SH_CLERK Shipping Clerk       2500     5500
8   SA_REP Sales Representative      6000    12008
7   SA_MAN   Sales Manager     10000    20080
9   PU_MAN Purchasing Manager      8000    15000
10  PU_CLERK Purchasing Clerk       2500     5500
18  PR_REP Public Relations Representative      4500    10500
6 AC_ACCOUNT Public Accountant      4200     9000
14   IT_PROG Programmer         4000    10000
0  AD_PRES  President    20080    40000
16  MK_REP Marketing Representative      4000     9000
15  MK_MAN Marketing Manager       9000    15000
17  HR_REP Human Resources Representative      4000     9000
3   FI_MGR   Finance Manager      8200    16000
1    AD_VP Administration Vice President  15000    30000
2  AD_ASST Administration Assistant      3000     6000
5   AC_MGR Accounting Manager      8200    16000
4 FI_ACCOUNT      Accountant      4200     9000
```